Sai Kiran

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.75
Narayana Co Spark Campus	Class XII (BIEAP)	2019	9.94%
Ravindra Bharathi School	Class X (BSEAP)	2017	9.80%

SKILLS

- **Programming :** Python, C
- **ML/DL :** PyTorch, TensorFlow, Scikit-learn, CNNs, Transformers, Mamba, SSL, RL
- **Computer Vision :** Semantic Segmentation, HSI, IQA, Multimodal Fusion, CLIP, LLMs, VLMs, VLAs
- **Autonomous Driving :** Radar Perception, BEV, Sensor Fusion, Free-Space Detection, Trajectory Prediction, MPC, PID, AEB
- **Tools :** ROS, ROS2, Linux, Git, Docker, SLURM, MATLAB

EXPERIENCE

Research Assistant, IIT Hyderabad

- Developed **HIQE**, a no-reference hyperspectral image quality assessment framework using 66 statistical features across spatial, spectral, and joint spectral-spatial domains; achieved 98.44% accuracy on natural and 99.42% on synthetic distortions.
- Proposed **RGB-HSI** semantic segmentation by extending SIGMA's shared VSSM/Mamba backbone with a learnable modality-alignment adapter; evaluated on 4 hyperspectral driving datasets, achieving 96.74 mIoU on HSI_Road and 80.42 mIoU in the HSI_DriveV2 ablation.
- Developed **RL-assisted MPC** using recurrent SAC to adaptively tune cost matrices Q and R; real-time experiments reported 40 ms inference and approximately 79.3% lateral-error reduction.
- Developed **radar-only BEV free-space estimation** with Doppler-decoupled encoding, LiDAR supervision, asymmetric convolutions, and ASPP; achieved 0.743 IoU, 0.657 RDM IoU, and 6.232 RDM MAE on nuScenes, with qualitative transfer to TiAND without fine-tuning.
- Contributed to **TiTAN, an end-to-end autonomous-driving framework** covering language annotation, supervised fine-tuning, RL post-training, trajectory prediction, and waypoint-based MPC execution; also implemented/evaluated TTC-based AEB with radar/LiDAR perception and PID-based braking.

Project Assistant, NMICPS TiHAN Foundation, IIT Hyderabad

- Built an automated UAV-based hyperspectral imaging pipeline for crop water-stress analysis, covering preprocessing, denoising, feature extraction, and inference; the work subsequently led to a **patent** filing (Patent No. 202441064638, Aug. 2024).

Research Intern, NMICPS TiHAN Foundation, IIT Hyderabad

- Investigated and evaluated hyperspectral image denoising methods for UAV-based vegetation imagery, contributing to research published at **IEEE WHISPERS 2023.**

PUBLICATIONS

1. **K. Sai Kiran***, et al., "SIGMA-Based RGB-Hyperspectral Fusion for Semantic Segmentation in Autonomous Driving," *IEEE International Conference on Image Processing **(ICIP)***, 2026. **Accepted.**
2. **K. Sai Kiran***, et al., "Radar BEV Free-Space Estimation for Autonomous Vehicles using Doppler-Decoupled Encoding," *IEEE 104th Vehicular Technology Conference (**VTC-Fall**)*, 2026. **Accepted.**
3. **K. Sai Kiran***, et al., "TiQE - Text-to-AI Generated Image Quality Evaluator," *10th International Conference on Computer Vision and Image Processing (**CVIP**)*, 2025. **Accepted.**
4. A. U. G. Sankararao, **K. Sai Kiran**, et al., "Self-Supervised Learning for Hyperspectral Image Classification Using Joint-Embedding Predictive Architecture," *IEEE India Geoscience and Remote Sensing Symposium (**InGARSS**)*, 2026. **Accepted.**
5. A. U. G. Sankararao, **K. Sai Kiran**, et al., "Stover Nitrogen Estimation in Groundnut Using Hyperspectral UAV Imagery and Machine Learning," *IEEE Applied Sensing Conference (**APSCON**)*, 2025.

6. A. U. G. Sankararao, **K. Sai Kiran**, et al., "UAV-Borne Hyperspectral Imaging Dataset of Pearl Millet Canopy Water Stress," *International Conference on Agriculture-Centric Computation (**ICA**)*, Springer, 2024.
7. A. U. G. Sankararao, **K. Sai Kiran**, et al., "Hyperspectral Image Denoising: A Comparative Study on UAV-Based Vegetation Data," *IEEE Workshop on Hyperspectral Imaging and Signal Processing: Evolution in Remote Sensing (**WHISPERS**)*, 2023.

PROJECTS

TiTAN - End-to-End Autonomous Driving Platform

- Contributed to an end-to-end autonomous-driving stack spanning four stages- language annotation, supervised fine-tuning, RL post-training, and trajectory execution using TiAND data from 52 Indian cities and TiCRAD's 158 safety-critical scenarios.

HI-JEPA - Self-Supervised Hyperspectral Representation Learning

- Designed a self-supervised JEPA framework for hyperspectral representation learning with spectral-spatial tokenization and dual-domain masking, achieving up to 99.54% SVM OA across five HSI datasets with only 10% labeled samples.

Towards Robust Video Anomaly Detection Using Multi-Task Learning

- Integrated self-supervised classification and contrastive-learning components with frame prediction in a multi-task video anomaly detection framework, evaluated on UCSD Ped1 using a 3D-ResNet encoder and temporal score fusion.

TiQE - Text-to-AI Generated Image Quality Evaluator

- Developed a CLIP ViT-B/32 multimodal evaluator using full-image and 64 patch-level embeddings for coarse/fine-grained perception and text-image alignment; achieved 0.8948 SRCC for perception and 0.7766 SRCC for alignment on AGIQA-3K.

Robust Convex Formulations for Sparse Recovery under Heavy-Tailed Noise

- Implemented Huber-AMP for robust mmWave radar sparse recovery under Cauchy-distributed multipath outliers, reducing MSE by ~89% and NMSE by ~89% versus the l2 baseline while improving RSNR from 9.3453 to 0.3532 dB.

PATENT

Automated Pipeline Device for Analysis and Inference of UAV-Based Hyperspectral Images of Crop Water Stress. Patent No: 202441064638, **Filed** on Aug 27, 2024.

LEADERSHIP & ENGAGEMENT

Mentor, Prayas - IIT Hyderabad

- Mentored student club domains and proposed initiatives such as Prerna and Srusti to support school students' development.

Placement Coordinator, Office of Career Services - IIT Hyderabad

- Coordinated with companies for campus placement drives and served as a recruiter point of contact.

College Student President, MVGR College of Engineering

- Reinitiated student activities in the college and organized the 25th Annual Day celebration.

Department President, Department of ECE, MVGR College of Engineering

- Organized IoT Project Expos and IEEE SPACE event for the department.

Team Lead, FYFP - MVGR

- Organized initiatives supporting government-school students and proposed Project Setu to strengthen student-alumni engagement.

HOBBIES & INTERESTS

- Running · Fitness Training · Movies · Meme Creation